Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Quadratic Equations: Solving by Factorisation

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Unit 1: Quadratic Equations: Solving by Factorisation

Learning Objectives

• Students will recognise a quadratic equation in standard form $ax^2 + bx + c = 0$.

• Students will factorise quadratic expressions by splitting the middle term.

• Students will solve quadratic equations by factorisation and check each root.

Key Concepts

• A quadratic equation has degree 2; its standard form is $ax^2 + bx + c = 0$, where a is not zero.

• Factorisation rewrites a quadratic as a product of two linear factors, for example $x^2 + 5x + 6 = (x + 2)(x + 3)$.

• If a product is zero, at least one factor is zero — so $(x + 2)(x + 3) = 0$ gives $x = -2$ or $x = -3$.

• Splitting the middle term means finding two numbers whose product is $a \times c$ and whose sum is b.

Teaching Methodology (Step by Step)

1. Ask students to expand $(x + 2)(x + 3)$ and notice the link between the factors and the middle term.

2. Introduce standard form and the zero-product rule with one worked example on the board.

3. Model splitting the middle term for $x^2 + 7x + 12$, then solve it step by step.

4. Students solve four equations in pairs, starting with $a = 1$ and moving to $a \neq 1$.

5. Students check their roots by substituting them back into the original equation.

Classroom Activities

• Factor race: pairs factorise five expressions against the clock.

• Error hunt: students find and correct the wrong step in a worked solution.

• Root check: students substitute roots back and explain why both roots work.

Assessment Questions

1. Factorise $x^2 + 9x + 20$.

2. Solve $x^2 - 3x - 10 = 0$ by factorisation.

3. Why does $(x - 4)(x + 1) = 0$ give two different values of x?

Homework

• Solve $x^2 + 6x + 8 = 0$ and $2x^2 + 5x - 3 = 0$ by factorisation.

• Write one quadratic equation whose roots are 3 and -5, and show its factorised form.

Time Allocation

40 minutes